

§11.8 Power Series

Homework : 3,7,13,17,23,27,30,31,35,39,40

Definition : $\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1 (x-a) + c_2 (x-a)^2 + \dots$ is called power series

about a (or centered at a) (or in $(x-a)$)

Question : For what values of x do the corresponding series converge?

$$\bullet \quad \begin{array}{c} \text{---} \quad | \quad \text{---} \quad | \quad \text{---} \quad | \quad \text{---} \\ \quad \quad r-a \quad \quad \quad a \quad \quad \quad r+a \end{array}$$

$\Rightarrow$ 愈靠近 a (中心), 愈可能收敛。

- 收敛半径 (radius of convergence)
- 收敛区间 (interval of convergence)

Example 1 : $\sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$

Solution : Using ratio test, we get

$$\lim_{n \rightarrow \infty} \frac{\left| \frac{(x-3)^{n+1}}{n+1} \right|}{\left| \frac{(x-3)^n}{n} \right|} = \lim_{n \rightarrow \infty} \frac{n|x-3|}{n+1} = |x-3|.$$

If $|x-3| < 1$, that is $x \in (2, 4)$, the power series converges

(absolutely).

- Check end points:

$$x=2 \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \Rightarrow \text{convergence.}$$

$$x=4 \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n} \Rightarrow \text{divergence.}$$

$\Rightarrow$ interval of convergence : $[2, 4)$.

radius of convergence : $r = 1$.

Example 2 : $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n} (n!)^2}$

Solution :

$$\begin{aligned}\text{Ratio test } \Rightarrow \lim_{n \rightarrow \infty} \frac{\left| \frac{x^{2(n+1)}}{2^{2(n+1)} ((n+1)!)^2} \right|}{\left| \frac{x^{2n}}{2^{2n} (n!)^2} \right|} \\ \Rightarrow \lim_{n \rightarrow \infty} \frac{x^2}{4(n+1)^2} = 0 \text{ for any } x \in R.\end{aligned}$$

$\Rightarrow$ interval of convergence: $(-\infty, \infty)$

$\Rightarrow$ radius of convergence: $r = \infty$.

註：重量級在分母 $(n!)^2 \gg x^{2n}$.

Example 3 : $\sum_{n=1}^{\infty} n! x^n$

Solution :

$$\text{Ratio test } \Rightarrow \lim_{n \rightarrow \infty} (n+1)x = \text{不存在 for any } x \in R.$$

$\Rightarrow$ interval of convergence: $\{0\}$.

$\Rightarrow$ radius of convergence: $r = 0$.

註：重量級 $(n! \gg x^n)$ 在分子.

Example 4 : $\sum_{n=1}^{\infty} \frac{2^n (x-1)^n}{n}$

Solution :

$$\text{Ratio test } \Rightarrow \lim_{n \rightarrow \infty} \left| \frac{2(x-1)n}{n+1} \right| = 2|x-1| \text{ for any } x \in R.$$

Let $2|x-1| < 1 \Rightarrow |x-1| < \frac{1}{2} \Rightarrow x \in \left(\frac{1}{2}, \frac{3}{2}\right)$.

- Check end points:

$$x = \frac{1}{2} \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \Rightarrow \text{convergence.}$$

$$x = \frac{3}{2} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n} \Rightarrow \text{divergence.}$$

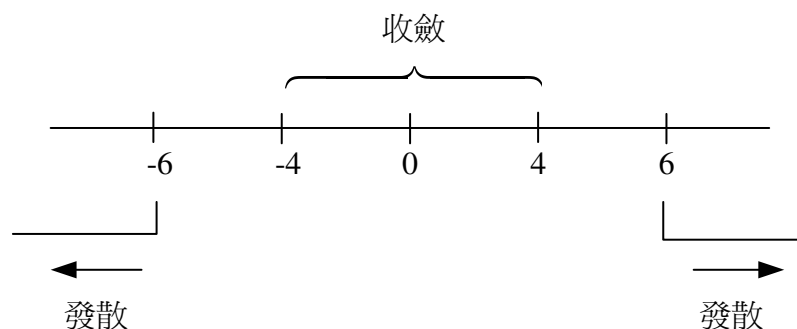
$$\Rightarrow \text{interval of convergence : } x \in \left[\frac{1}{2}, \frac{3}{2}\right).$$

$$\text{radius of convergence : } r = \frac{1}{2}.$$

Example 5 : If $\sum_{n=1}^{\infty} c_n x^n$ converges when $x=4$ and diverges when $x=6$, then determine the convergence of the following series.

	answers	reasons
i. $\sum_{n=1}^{\infty} c_n$	convergence	$x=1$
ii. $\sum_{n=1}^{\infty} c_n 8^n$	divergence	$x=8$
iii. $\sum_{n=1}^{\infty} c_n (-3)^n$	convergence	$x=-3$
iv. $\sum_{n=1}^{\infty} (-1)^n c_n 9^n$	divergence	$x=9$

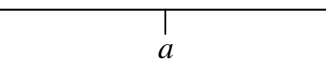
由已知可得下圖：

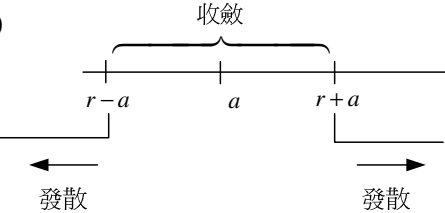


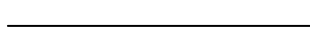
註：If $4 \leq |x| \leq 6$, then 無從判斷收斂性。

Theorem :

$\sum_{n=1}^{\infty} c_n (x-a)^n$, then either one of the following 3 possibilities holds :

(i)  , $r=0$ (只有一點收斂)。

(ii)  , $r < \infty$ (兩端點需被檢驗才可知其收斂與否)。

(iii)  , $r = \infty$ (每一點皆收斂)。